

Module 4 - Readings

This week's reading assignment includes the following:

- **Textbook:** Nair, M. T., Singh, A. (2019). *Linear Algebra*. Singapore: Springer Singapore.
 - Chapter 2, Sections 2.4-2.5

The reading may be accessed through EReserves.

Setup

For $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$, $T(a, b, c) = (b + c, c + a, a + b)$, the matrix $[T]_{C,B}$ has as its j -th column the **coordinate vector of $T(b_j)$ relative to C** .

(a) B standard, $C = \{(1,0,0), (1,1,0), (1,1,1)\}$

First find T on the standard basis vectors:

$$T(1, 0, 0) = (0, 1, 1), \quad T(0, 1, 0) = (1, 0, 1), \quad T(0, 0, 1) = (1, 1, 0)$$

For $C = \{c_1, c_2, c_3\} = \{(1, 0, 0), (1, 1, 0), (1, 1, 1)\}$, a vector (x, y, z) has C -coordinates

$$(x - y, y - z, z)$$

(since $\alpha c_1 + \beta c_2 + \gamma c_3 = (\alpha + \beta + \gamma, \beta + \gamma, \gamma)$).

Applying this to each image:

- $(0, 1, 1) \rightarrow (-1, 0, 1)$
- $(1, 0, 1) \rightarrow (1, -1, 1)$
- $(1, 1, 0) \rightarrow (0, 1, 0)$

$$[T]_{C,B} = \begin{pmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 1 & 1 & 0 \end{pmatrix}$$

(b) $B = \{(1,0,0), (1,1,0), (1,1,1)\}$, C standard

Compute T on the basis vectors of B :

$$T(1, 0, 0) = (0, 1, 1)$$

$$T(1, 1, 1) = (2, 2, 2)$$

Since C is standard, coordinates = components:

$$[T]_{C,B} = \begin{pmatrix} 0 & 1 & 2 \\ 1 & 1 & 2 \\ 1 & 2 & 2 \end{pmatrix}$$

$$(c) B = \{(1,1,2), (2,1,1), (1,2,1)\}, C = \{(2,1,1), (1,2,1), (1,1,2)\}$$

Images of B-vectors under T:

$$T(1, 1, 2) = (3, 3, 2), \quad T(2, 1, 1) = (2, 3, 3), \quad T(1, 2, 1) = (3, 2, 3)$$

Express each in terms of C : Let $M = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}$ (columns are C -vectors). Then $\det M =$

4 and

$$M^{-1} = \frac{1}{4} \begin{pmatrix} 3 & -1 & -1 \\ -1 & 3 & -1 \\ -1 & -1 & 3 \end{pmatrix}$$

Compute coordinates $M^{-1}v$:

- For $v = (3, 3, 2)$: $\left(\frac{9-3-2}{4}, \frac{-3+9-2}{4}, \frac{-3-3+6}{4}\right) = (1, 1, 0)$
- For $v = (2, 3, 3)$: $\left(\frac{6-3-3}{4}, \frac{-2+9-3}{4}, \frac{-2-3+9}{4}\right) = (0, 1, 1)$
- For $v = (3, 2, 3)$: $\left(\frac{9-2-3}{4}, \frac{-3+6-3}{4}, \frac{-3-2+9}{4}\right) = (1, 0, 1)$

(Quick check: $M(1, 1, 0)^T = (3, 3, 2) \checkmark$, $M(0, 1, 1)^T = (2, 3, 3) \checkmark$, $M(1, 0, 1)^T = (3, 2, 3) \checkmark$.)

$$[T]_{C,B} = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}$$

This is **Exercise 1 of Section 2.4 (Matrix Representation)**, found on p. 84 of *Nair & Singh, Linear Algebra* (Springer, 2018). The theory needed is all in Section 2.4, pp. 76–83:

- **Definition 2.37** (p. 76) — coordinate vector $[x]_B$ of x relative to an ordered basis B .
- **Definition 2.38** (p. 78) — the matrix representation $[T]_{C,B}$: if $B = \{v_1, \dots, v_n\}$, $C = \{w_1, \dots, w_m\}$, and

$$T(v_j) = a_{1j}w_1 + \dots + a_{mj}w_m,$$

then $[T]_{C,B} = [a_{ij}] \in \mathbb{F}^{m \times n}$.

- **Key rule** (derived right after Theorem 2.41, p. 80): *the j -th column of $[T]_{C,B}$ is $[T(v_j)]_{C^*}$ — the coordinate vector, w.r.t. C , of T applied to the j -th vector of B .
- **Theorem 2.41** (pp. 79–80): $[Tx]_C = [T]_{C,B}[x]_B$.
- The worked **Example 2.39** (p. 77–78) shows exactly this procedure for a map $\mathbb{R}^2 \rightarrow \mathbb{R}^3$.

So the algorithm is always the same **two-step process**:

1. Compute $T(v_j)$ for every vector v_j in the domain basis B .
2. Write $T(v_j)$ as a linear combination of the vectors of the codomain basis C — the coefficients become **column j** of $[T]_{C,B}$.

Here $T(a, b, c) = (b + c, c + a, a + b)$ throughout.

(a) $B = \{e_1, e_2, e_3\}$ (standard), $C = \{(1, 0, 0), (1, 1, 0), (1, 1, 1)\}$

Step 1 — images of the B -vectors:

$$T(1, 0, 0) = (0, 1, 1), \quad T(0, 1, 0) = (1, 0, 1), \quad T(0, 0, 1) = (1, 1, 0)$$

Step 2 — express each image in terms of $C = \{c_1, c_2, c_3\} = \{(1, 0, 0), (1, 1, 0), (1, 1, 1)\}$.

If $xc_1 + yc_2 + zc_3 = (p, q, r)$, then

$$xc_1 + yc_2 + zc_3 = (x + y + z, y + z, z) = (p, q, r)$$

Reading off from the bottom equation upward (this is exactly the "express v as a combination of basis vectors" step used in Example 2.39):

$$z = r, \quad y = q - r, \quad x = p - q.$$

Apply this formula $(p, q, r) \mapsto (p - q, q - r, r)$ to each image:

image (p, q, r)	coordinates (x, y, z) w.r.t. C
$(0, 1, 1)$	$(-1, 0, 1)$
$(1, 0, 1)$	$(1, -1, 1)$
$(1, 1, 0)$	$(0, 1, 0)$

By Definition 2.38, these coordinate vectors are the **columns** of $[T]_{C,B}$:

$$[T]_{C,B} = \begin{pmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 1 & 1 & 0 \end{pmatrix}$$

Check (Theorem 2.41, $[Tx]_C = [T]_{C,B}[x]_B$): for $x = e_1$, $[x]_B = (1, 0, 0)^T$, and $[T]_{C,B}(1, 0, 0)^T = (-1, 0, 1)^T$ — matches $T(e_1) = (0, 1, 1) = -1 \cdot c_1 + 0 \cdot c_2 + 1 \cdot c_3$. ✓

(b) $B = \{(1, 0, 0), (1, 1, 0), (1, 1, 1)\}$, $C = \{e_1, e_2, e_3\}$ (**standard**)

Step 1 — images of the B -vectors (plug (a, b, c) into $T(a, b, c) = (b + c, c + a, a + b)$):

$$T(1, 0, 0) = (0, 1, 1)$$

$$T(1, 1, 1) = (2, 2, 2)$$

Step 2 — express in terms of C : since C is the standard basis, coordinates equal components directly (no work needed — this is stated right after Theorem 2.42, p. 81: " T and $[T]_{C,B}$ coincide when B and C are the standard bases").

Columns = these images themselves:

$$[T]_{C,B} = \begin{pmatrix} 0 & 1 & 2 \\ 1 & 1 & 2 \\ 1 & 2 & 2 \end{pmatrix}$$

$$(c) B = \{(1, 1, 2), (2, 1, 1), (1, 2, 1)\}, C = \{(2, 1, 1), (1, 2, 1), (1, 1, 2)\}$$

Step 1 — images of the B -vectors:

$$T(1, 1, 2) = (1 + 2, 2 + 1, 1 + 1) = (3, 3, 2)$$

$$T(2, 1, 1) = (1 + 1, 1 + 2, 2 + 1) = (2, 3, 3)$$

$$T(1, 2, 1) = (2 + 1, 1 + 1, 1 + 2) = (3, 2, 3)$$

Step 2 — express each image in terms of $C = \{c_1, c_2, c_3\} = \{(2, 1, 1), (1, 2, 1), (1, 1, 2)\}$.

Solve $xc_1 + yc_2 + zc_3 = (2x + y + z, x + 2y + z, x + y + 2z) = (p, q, r)$.

Trick: let $s = x + y + z$. Each coordinate equals $s +$ (that variable), so

$$s + x = p, \quad s + y = q, \quad s + z = r \implies 3s + s = p + q + r \implies s = \frac{p+q+r}{4}.$$

Then $x = p - s$, $y = q - s$, $z = r - s$.

(This is equivalent to inverting the matrix $M = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}$ whose columns are c_1, c_2, c_3 ;

$$\det M = 4 \text{ and } M^{-1} = \frac{1}{4} \begin{pmatrix} 3 & -1 & -1 \\ -1 & 3 & -1 \\ -1 & -1 & 3 \end{pmatrix}.)$$

- $T(v_1) = (3, 3, 2): p + q + r = 8 \Rightarrow s = 2; (x, y, z) = (1, 1, 0)$
- $T(v_2) = (2, 3, 3): p + q + r = 8 \Rightarrow s = 2; (x, y, z) = (0, 1, 1)$
- $T(v_3) = (3, 2, 3): p + q + r = 8 \Rightarrow s = 2; (x, y, z) = (1, 0, 1)$

Verification: $1 \cdot c_1 + 1 \cdot c_2 + 0 \cdot c_3 = (2, 1, 1) + (1, 2, 1) = (3, 3, 2) = T(v_1) \checkmark$ (similarly for the others).

Columns give:

$$[T]_{C,B} = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}$$

Summary of what was used

- **Definition 2.38** (p. 78): definition of $[T]_{C,B}$ via $T(v_j) = \sum_i a_{ij} w_i$.
- Rule following **Theorem 2.41** (p. 80): j -th column of $[T]_{C,B}$ is $[T(v_j)]_C$.
- **Theorem 2.41** itself (pp. 79–80): $[Tx]_C = [T]_{C,B}[x]_B$, used for verification.
- Method modeled on **Example 2.39** (pp. 77–78), the direct analogue in the book to this exact exercise.

Assignment 4

Assignments will consist of problems from the textbook and additional problems that require the use of statistical software. All assignment submissions should be saved as a PDF file that includes the module number and your last name (ex. mod1_bodt.pdf). Submit the assignment by clicking on the blue 'Start Assignment' button. Assignments are due according to the dates on the course calendar.

Part 1: Problems

3.10.4, 3.10.6, 3.10.9, 3.10.14, 3.12.4, 3.12.8, 3.12.14, 3.12.16, 3.12.22

Part 2: Software Assignment

We will confirm your results in the HW for problems 3.10.4 and 3.10.6. To do this, you can use the Inverse Transform Sampling Method for random number generation from an arbitrary continuous distribution having an inverse. For a continuous random variable, X , having distribution $F_X(x)$, it is possible to generate random deviates of X by forming $F_X^{-1}(u)$, where u is a random number from a uniform distribution on $[0,1]$.

MC 3.10.4

Find the cdf for the density in problem 3.10.4. Use that to randomly generate a sample of size 5, call it y . Sort the sample and determine if the interval Y'_1 to Y'_5 contains 0.60. The interval is $y[1]$ to $y[5]$. Repeat for 1000 trials and compare with your theoretical result.

MC 3.10.6

After determining n analytically from problem 3.10.6, sample n values repeatedly from the distribution of Y . You will use the approach above, first finding the distribution function and then using the inverse distribution evaluated at a random uniform value on $[0,1]$ to generate a random deviate for Y . For each of the repetitions of n values, sort the y values and identify the minimum $y[1]$ for comparison to 0.2. Compute the sample proportion in 1000 trials of n values to see if the $P(Y_{\min} < 0.2) > 0.9$. If your computed n is correct, the probability should be very close to 0.9.

This assignment is due by Day 7 of Module 4.